

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and executed **small-scale protein purification studies** using AKTA systems, analyzing results to support process development for enzyme and antibody targets, with limited supervision.
- Authored comprehensive **SOPs, risk assessments, and technical reports** for protein purification workflows, ensuring adherence to Good Laboratory Practices (GLP) and ICH guidelines.
- Led root cause analysis investigations for processing issues in small-scale protein purification, providing technical guidance to junior scientists and developing their expertise.
- Developed and maintained **small-scale models and expertise** in laboratory protein purification techniques, including chromatography resin screening and filtration methods.
- Collaborated with cross-functional teams including Process Development, Manufacturing, and Analytical Services to define project plans for R&D initiatives.
- Contributed to a culture of **Operational Excellence** by implementing standard work and 5S principles within the laboratory environment.
- Containerized Python-based data processing scripts using **Docker** and deployed them to **AWS Lambda** for efficient execution of analytical tasks.

SKILLS

Bioprocessing – Small-scale protein purification, AKTA systems, TFF, Filtration, Chromatography, Resin screening, Enzymes, Antibodies

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, R, RStudio, pandas, NumPy, scikit-learn

Programming & Scripting – Python, Java, Spring Boot, REST APIs, SQL, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Core Technologies – PostgreSQL, MySQL, Oracle, Linux, Jira, Confluence

Quality & Compliance – GMP, ICH guidelines, GLP, SOPs, Risk Assessment

PROJECTS

Protein Purification Process Development (HCL Technologies)

- Designed and performed **small-scale chromatography** experiments (e.g., ion exchange, affinity) to optimize buffer conditions and resin selection for a target therapeutic antibody.
- Developed **data analysis pipelines in Python** using pandas to process raw output from AKTA systems, calculating key purification metrics like yield and purity.
- Generated a technical report detailing experimental findings, including recommendations for scaling up the purification process, for review by senior R&D staff.

Analytical Data Pipeline for Batch Release (MSc Project)

- Built a **Python-based ETL pipeline** to extract, transform, and load data from simulated manufacturing batch records into a PostgreSQL data warehouse.
- Developed **Power BI dashboards** to visualize key batch quality attributes and process performance indicators, enabling faster review by hypothetical Quality Assurance stakeholders.
- Applied **statistical analysis in R** to identify correlations between process parameters and product quality, informing potential areas for continuous improvement.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)